Tutorial 4

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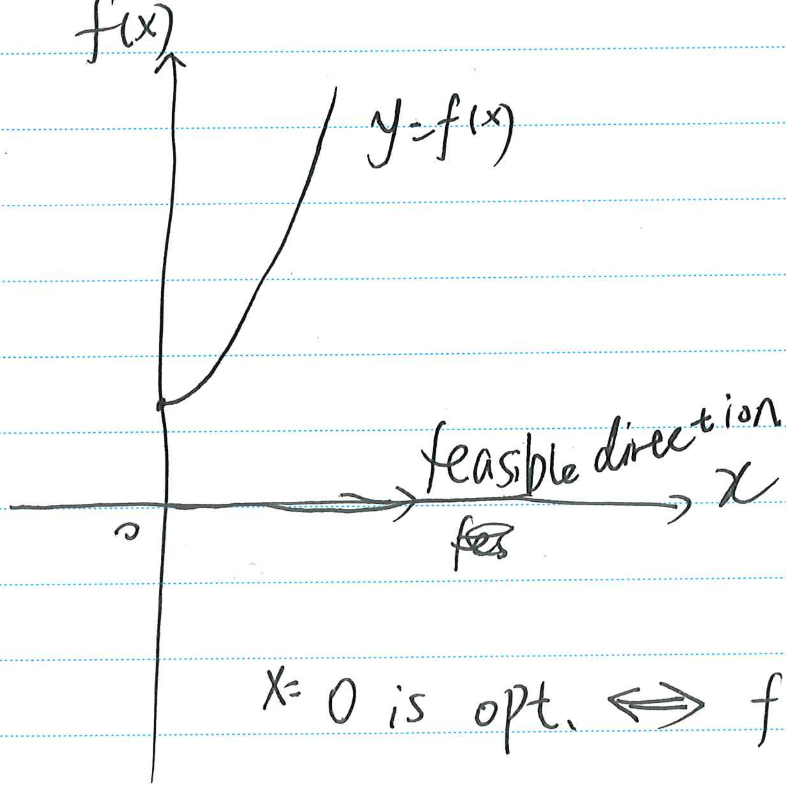
Opt. criterion for differentiable fo

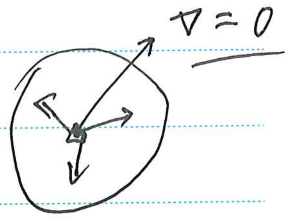
$\min f(x)$ $x \in R$

~~s.t.~~ s.t. $-x \leq 0$

~~$x \leq 0$~~ (i)

Feasible Region $\{x \mid x \geq 0\}$ (only decided by constraints)



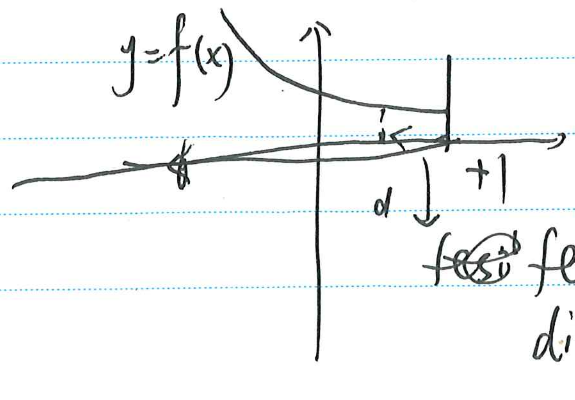


$x=0$ is opt. $\Longleftrightarrow$ $f'(0) \geq 0$

<0 $\lim\limits_{x \to 1}$

(ii)

Feasible Reigon $= (-\infty, +1]$



~~x~~ $x=1$ is opt.

$\Leftrightarrow$ $f'(1) \leq 0$

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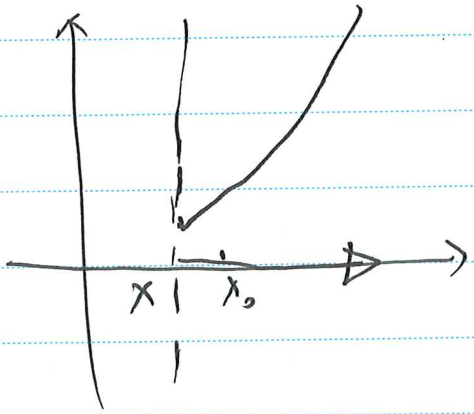

$f'(x) \geq 0$
$\Leftrightarrow$ for a feasible direction at x^0
~~for any feasible direction~~

In 1D case:

x^0 is opt.
$\Leftrightarrow$ $f'(x^0) \cdot d \geq 0$

for any feasible direction at x^0

$\nabla f(x)^T \cdot (y-x) \geq 0$

$d = y - x^*$ ~~fesi~~ feasible direction at x^*

定理
$\nabla f(x^*)^T \cdot d \geq 0$ for any direction at x^*

Tutorial 5

Problem 1

If we would like to prove x^* is ~~a~~ optimal
We just need to prove that $f(x) = (\frac{1}{2})x^TPx + q^Tx + r$
is meets the requirement that for any d, there will be always $\nabla f(x^*)d > 0$

∴ That means for any d, there will be a linear combination $A = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_n \end{pmatrix}$.

Moreover, ∵ $\nabla f(x^*) = (a_1, a_2, a_3 \cdots\cdots, a_n) \cdot \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \vdots \\ \lambda_n \end{pmatrix}$

$= A^T - (v) = -A^Tv$

∵ x^* is optimal for a convex problem
∴ x^* is feasible and ~~$\nabla f(x^*)$~~ $\nabla f(x^*)^T(y - x^*) \geq 0$

∴ for any $y \in$ feasible region.
$\nabla f(x^*) = Px^* + q = \cancel{(\tfrac{1}{2})} \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$

Moreover, $\because \begin{pmatrix}-1\\0\\2\end{pmatrix}\begin{pmatrix}y_1-1\\y_2-\frac{1}{2}\\y_3+1\end{pmatrix} > 0$ is feasible

$\therefore$ $\cancel{(1-y_1)+2(}$ $(1-y_1)+2(y_3+1)\geq 0$
for any feasible y

$\therefore x^* = (+1, -\frac{1}{2}, -1)$ is optimal for the optimization problem.

Problem 2

(b) This problem is always fesible.
The vector $\vec{c}$ can be decomposed into a component parallel to a and a component orthogonal to a:

$$c = a\lambda + \hat{c} \quad (\text{with } a^T\hat{c} = 0)$$

(i) if ~~λ=0~~ $\lambda > 0$:
Now, the problem is ~~always~~ unbounded below.
We can choose $x = -ta$, and let t go to infinity:

$$c^T x = -tc^T a = -t\lambda a^T a \to -\infty$$

and

$$a^T x - b = -ta^T a - b \le 0 \text{ (for large } t)$$

~~For large t~~

so x is fesible for large t. Intuitively, by going very far in the direction $-a$, we find fesible points with arbitrarily negative objective values.

(ii) if $\hat{c} \neq 0$:

The problem is unbounded ~~beto~~ below.

we choose $x = ba - t\hat{c}$ and let t go to infinity.

(iii) if $c \cancel{\neq} = a\lambda$ for some $\lambda < 0$:

the optimal value is $c^T ab = \lambda b$

In summary, the ~~opit~~ optimal value is:

$$p^* = \begin{cases} \lambda b, & c = a\lambda \text{ for some } \lambda \leq 0 \\ -\infty, & \text{otherwise} \end{cases}$$

Lecture 6

(1) $f(x)$ is given

$x \in R^n$

$\therefore \; \underset{\text{convex}}{f^*(y)} \triangleq \sup\limits_{x \in \operatorname{dom} f} \{ \underbrace{x^T y - f(x)}_{\text{affine function on } y} \}$

~~for~~ on y

$f(x) + f^*(x) \geq x^T y$

If $f(x)$ is convex $\operatorname{dom} f = R^n$

Then $f(x) = f^{**}(x)$

证明: 设 $f^*(y) = h(y)$

$f(x) = h^*(x) = \sup\limits_{y} \{ x^T y - h(y) \}$

affine function

f in X

(2) when:

~~(2)~~ $f(x) \in C^1$

f is convex $\Leftrightarrow \forall x, y$

$f(y) \geq f(x) + \nabla f(x)^T (y - x)$

(3) 琴生不等式 (Jensen's inequality)

Assume that $x^1, \cdots\cdots, x^k \in R^n$

$\theta_1, \cdots, \theta_k \geq 0$, $\sum_{i=1}^{k} \theta_i = 1$

$\therefore \; f\left(\sum_{i=1}^{k} \theta_i x^i\right) \leq \sum_{i=1}^{k} \theta_i f(x^i)$

$f(Ex) \leq Ef(x)$

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$\tilde{x}$ is feasible, $\lambda \geq 0$

$f_0(x) \geq L_1(\tilde{x}, \lambda, v)$

minimizing over all feasible $\tilde{x}$

gives $p^* \geq g(\lambda, v)$

范数?

$x^T x = x_1^2 + x_2^2 + x_3^2 + \cdots + x_n^2$

$$Ax = b \Leftrightarrow \begin{cases} a_i^T x_i - b_i = 0 \\ \vdots \\ a_m^T x - b_m = 0 \end{cases}$$

线性相关?

$L(x, v) = x^T x + v_1(a_1^T x - b_1) + \cdots + v_m(a_m^T x - b_m)$

$= x^T x + \cancel{v^T(ax-b)}\ v^T(Ax - b)$

即:

$$(v_1, \cdots, v_m) \begin{pmatrix} a_1^T x - b_1 \\ \vdots \\ a_m^T x - b_m \end{pmatrix}$$

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Why we ~~can~~ can't do max-min together?

$f(x) = c^T x$

$$Ax = b \iff \begin{cases} a_1^T x - b_1 = 0 \\ \vdots \\ a_m^T x - b_m = 0 \end{cases} \quad b = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix} \quad A = \begin{pmatrix} a_1^T \\ \vdots \\ a_m^T \end{pmatrix}$$

$$x \geq 0 \iff \begin{cases} -x_1 \leq 0 & f_1(x) = -x_1 \\ \vdots & \\ \vdots & \\ -x_m \leq 0 & f_m(x) = -x_m \end{cases}$$

$$L(x, \lambda, v) = c^T x + \lambda_1 \cdot (-x_1) + \lambda_2(-x_2) + \cdots + \lambda_m(-x_m) + v_1(a_1^T x - b_1) + \cdots + v_m(a_m^T x - b_m)$$

$$= c^T x + v^T(Ax - b) - \lambda^T x$$
$$= -b^T v + (c + A^T v - \lambda)^T x$$

lower bound property: $p^* \geq b^T v$

if $A^T v + \cancel{0}c \geq 0$

Constraints: $\begin{cases} \lambda \geq 0 \\ A^T v - \lambda + c = 0 \end{cases} \Rightarrow A^T v + c \geq 0$

Actually: d^* $\left| \begin{array}{l} \max = -b^T v \\ \text{s.t. } A^T v + c \geq 0 \end{array} \right.$

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conjugate: $f^*(y) = \sup_x \{x^T y - f(x)\}$

$\therefore \inf\{-x^T y + f(x)\}$

$= -\sup_x \{-(-x^T y + f(x))\}$

$= -\sup_x \{x^T y - f(x)\}$

$= -f^*(y)$

※ Dual Problem is one kind of maxmization problem

Constraint qualifications ⇔ 约束品格

若满足，则 $p^* = d^*$

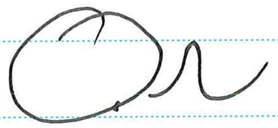
Lagrangian and dual function

p^*

$\min f_0(x)$

s.t. $x \in \Omega$

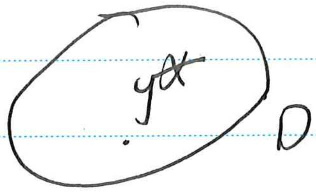
$f_0(x) \geq g(y)$



$\max \cancel{g(y)}\ g(y)$

d^* s.t. $y \in D$

$\therefore p^* \geq d^*$



d^* is a good lower bound for p^*

$\cancel{d^*}$

$L(x, \lambda, \nu) = f_0(x) + \lambda_1 f_1(x) + \lambda_2 f_2(x) + \nu_1 h_1(x)$

$$\underbrace{g(\lambda, \nu)}_{\text{concave}} = \inf_{x \in D} L(x, \lambda, \nu) \quad (\lambda \geq 0)$$

$$= -\sup_{x \in D} \{-L(x, \lambda, \nu)\}$$

Revision: inplict function

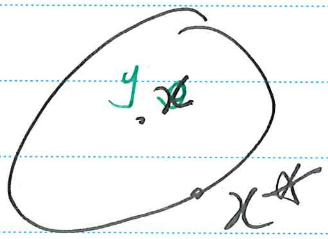
$\min f_0(x)$

~~fo~~ $f_i(x) \leq 0$

$h_i(x) = 0$

$x \in \frac{D}{J}$ inplict function

(4) First-order sufficient and necessary optimality condition



suppose x^* is OPTIMAL

$\Longleftrightarrow \forall$ ~~fesi~~ feasible y

iff $\nabla f(x^*)^T (y - x^*) \geq 0$

Mid-term exam's solution

T1
(1) Convex
(2) Concave
(3) Convex
(4) Concave
(5) Convex
(6) Concave
(7) Convex

T2
(1) convex
Definition
(2) conjugate
(3) hyperplane
(4) intersection of halfspaces:
$a(x_1-a)+b(x_2-b)\le 0$
for all $a^2+b^2=1$
(e) $x_1=2$

T3

(1)
$f^{*}(y) = \sup\{xy-(3x-4)\}$

$\therefore f^*(y) = \begin{cases} 4 & y=3 \\ +\infty & y\neq 3 \end{cases}$

(2) $f^*(y) = \sup(xy-(e^x))$

~~$f^*(y) = \{$~~

(2) $f^*(y) = \sup\{xy - e^x\}$

when $y<0$, $f^*(y) = +\infty$

when $y=0$, $f^*(y) = 0$

when $y>0$: let ~~hy~~ $h(x) = xy - e^x$

$\therefore h'(x) = y - e^x = 0 \therefore x = \ln y$

$\therefore f^*(y) = y\ln y - y$

T4 $\max\limits_{x\in S} f(x) = \max\limits_{1\le k\le m} f(V_k)$

proof: $f(x) \le \sum \lambda_i f(V_i) \le \max\limits_{x_k} f(V_k)$

since $V_k \in S$, so: $\max\limits_{x\in S} f(x) = \max\limits_{1\le k\le m} f(V_k)$

T5: (a) H(x) shows that it is Semi-Positive

$\therefore$ h(x) is Convex

(b) $F(\lambda X_1 + (1-\lambda) X_2) = \dfrac{[f(\lambda x_1 + (1-\lambda) X_2)]^2}{g(\lambda_1 x_1 + (1-\lambda) X_2)}$

$\Rightarrow$ More $\because$ f is Convex
g is concave

$$F(\lambda X_1 + (1-\lambda)X_2)) \le \frac{[\lambda f(X_1) + (1-\lambda) f(X_2)]^2}{\lambda g(X_1) + (1-\lambda) g(X_2)}$$

$$= h(\lambda f(X_1) + (1-\lambda) X_2), \lambda g(X_1) + (1-\lambda) g(X_2))$$

$$\le \lambda h(f(X_1), g(X_1)) + (1-\lambda) h(f(X_2), g(X_2))$$

$$= \lambda F(X_1) + (1-\lambda) F(X_2)$$

New for Week ~~8~~ 98 Lecture 7

KKT condition

Assume: $\begin{cases} \text{i) Strong duality } (p^* = d^*) \\ \text{(ii) } x^* \text{ is primal optimal} \\ \text{(iii) } (\lambda^*, \mu^*) \text{ is dual optimal} \end{cases}$

$$f_0(x^*) = g(\lambda^*, \mu^*) = \inf_x \left(f_0(x) + \sum_{i=1}^{m} \lambda_i^* f_i(x) + \sum_{i=1}^{p} \mu_i^* h_i(x) \right)$$

$$\leq f_0(x^*) + \sum_{i=1}^{m} \lambda_i^* f_i(x^*) + \sum_{i=1}^{p} \mu_i^* h_i(x^*)$$

$$\leq f_0(x^*)$$

$$\begin{cases} \lambda_i^* > 0 \Rightarrow f_i(x^*) = 0 \\ f_i^*(x) < 0 \Rightarrow \lambda_i^* = 0 \end{cases}$$

KKT condition

$x^0, \lambda + \mu$

Given $\Rightarrow$ [4th condition]

$$\nabla_x L_1(x^0, \lambda^0, \mu^0) \Big|_{x = x^0} = 0$$

$$\Rightarrow \nabla f_0(x) + \sum_{i=1}^{m} \lambda_i \nabla f_i(x) + \sum_{i=1}^{m} \mu_i \nabla h_i(x) = 0$$

Review

①

$\min f(x)$

s.t. $f(x) \le 0$

primal $f_2(x) \le 0$

(P)

question $h_1(x) = 0$

if $h_2(x) = 0$

$x \in$ Domain all functions are defined.

Otherwise: there will be not.

②

Optimization

~~Opt~~ of P

$\Rightarrow$ is p^*

$L(x, \lambda, \mu)$

$\triangleq f_0(x) + \sum_{i=1}^{\infty} \lambda_i f_i(x) + \sum_{j=1}^{\infty} \mu_j h_j(x)$

$x \in D$, $\lambda \ge 0$ dual (d question)

Dual function

$g(\lambda, \mu) \triangleq \inf_{x \in D} \{L(x, \lambda, \mu)\}$

Con cave on $\lambda > 0$

③ $\max g(\lambda, \mu) \le \min L(x, \lambda, \mu)$

s.t. $\lambda \ge 0$ s.t. Convex

(d^*) $\lambda \ge 0$

If x is feasible to ~~P D~~ P
and $\lambda \geq 0$
Then: $f_0(x) \geq g(\lambda, \mu)$

Week duality: $p^* \geq d^*$ is always true

Strong duality: $p^* = d^*$ is always true
($p^* \triangleq d^*$)

For a convex problem (P),
~~slector's condition~~ is true
Slater's constraint qualification
Then: $p^* = d^*$

Perturbed Problem

$$f_0(x) + \lambda_1 (f_1(x) - u_1) + \mu_1 (h_1(x) - v_1)$$

Global sensitivity via duality

$$p^*(u,v) \geq g(\lambda^*, \mu^*) - u^T \lambda^* - v^T \mu^*$$

$$= p^*(0,0) - u^T \lambda^* - v^T \mu^*$$

Local sensitivity via duality

$$p^*(u,v) \geq p^*(0,0) - \lambda^* u - \mu^* v$$

Farkas' lemma

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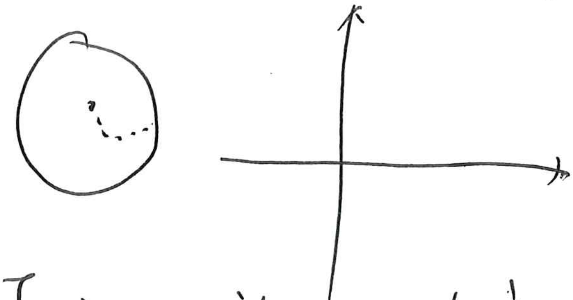
Lecture 8

Theory
Algorithms

$\nabla f(x)$
$\nabla^2 f(x) \triangleq H(x)$
D is open

$\exists\, x^*$
s.t. $f(x^*) = p^*$

$S = \{x \mid f(x) \leq f(x^0)\}$



if x^* is OPTIMAL
$\Leftrightarrow \nabla f(x^*) = 0$

$f(x) = (\frac{1}{2}) x^T P x + q^T x + r$, $I \Rightarrow$ unity matrix

$P \geqslant 0$

$\nabla^2 f(x) \succeq mI \Leftrightarrow$ every eigenvalue of $\nabla^2 f(x) \geq m > 0$

$\therefore \nabla f(x) = Px + q = 0$

Why?
$A, B \in S^n$:
$A \succeq B \Leftrightarrow A - B \succeq 0$

$\min f(x) \Leftrightarrow \nabla f(x) = 0$

$x^0 \to x^1 \to x^2 \to \cdots \to x^k$ $\Leftrightarrow \forall x \in R^n$,
$x^T (A - B) x \geq 0$

if $k \to +\infty$,

we hope $f(x^k) \to p^*$

f is convex $\Leftrightarrow$ at any x, $\nabla^2 f(x) \succeq 0$

$\Leftrightarrow$ we hope $\nabla f(x^*) \to 0$

For example:

$$g(x) = \frac{m}{2} x^T x$$

$$\therefore \nabla^2 g(x) = mI$$

$$\nabla^2 h(x) = \nabla^2 f(x) - mI \geq 0$$

explanation:

$\cancel{f(x) = f(x_i) + f'(x_i)(x}$

$\cancel{(x \in R)}$

$x \in R$

$$f(x) = f(x_i) + f'(x_i)(x - x_i) + \frac{1}{2} f''(\xi)(x - x_i)^2$$

$\therefore x \in R^n$

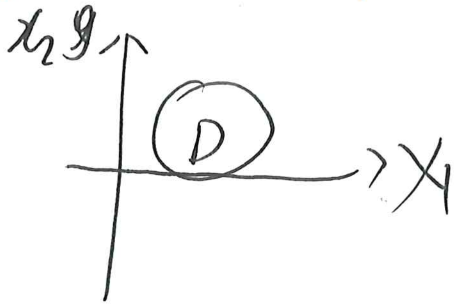
$$f(x) = f(x^0) + \nabla f(x^0)^T (x - x_0) + \frac{1}{2}(x - x_0)^T H(\xi)(x - x_0) + \cancel{\frac{\nabla^2 f(x^0)^T (x - x_0)^2}{2}}$$

Moreover: $\cancel{H(\xi) \geq mI}$

$H(\xi) \geq mI$

$$\therefore f(x) \geq f(x_0) + \nabla f(x - x_0)^T (x - x_0) + \frac{m}{2} \|x - x_0\|^2$$

D is bounded
(有界)



$\forall x$ GS

x_0 is starting point
not optimal point

$0 \le f(x) - p^* \le \frac{1}{2m} \|\nabla f(x)\|_2^2$

$x^k \to x^{k+1}$

$\Delta x^k \overset{?}{=} \mathcal{D}q$ (移动方向)

$t^k > 0$ (步长)

$\therefore x^{k+1} = x^k + t^k \Delta x^k$

$x^+ = x + t\Delta x$, $x := x + t\Delta x$

单点迭代法: $f(x^k) > f(x^{k+1})$

$f(x^+) < f(x) \Longrightarrow$

$\nabla f(x) \Delta x < 0$

回忆:

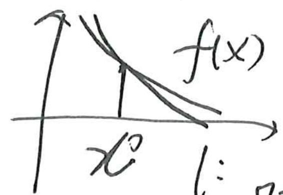
$f(x)$ is convex

$\Updownarrow$

$\forall x^0, x,$

$f(x) \ge f(x_0) + \nabla f(x_0)^T (x - x_0)$

$\Rightarrow$



$l: \nabla f(x_0)^T = \frac{f(x) - f(x_0)}{x - x_0}$

$[t\Delta x = x^+ - x]$

$f(x + t\Delta x) \ge f(x) + \nabla f(x)^T \cancel{(x^+ - x)} (x^+ - x)$

$\therefore f(x + t\Delta x) \ge f(x) + t \nabla f(x)^T \Delta x$

$\therefore \nabla f(x) \Delta x < 0$

directional derivatives

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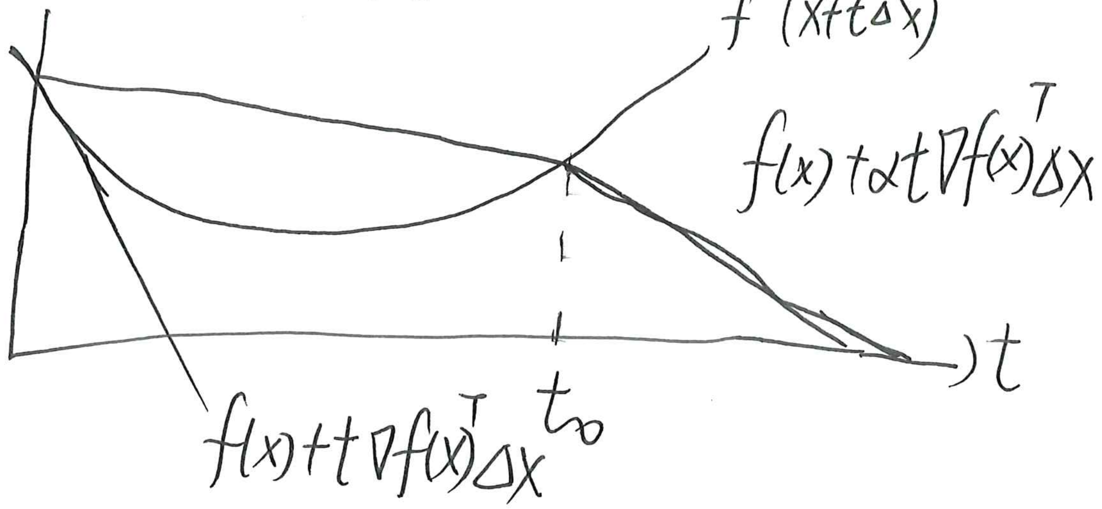
How do we get a great $t/\Delta x$?

starting at $t=1$, repeat $t := \beta t$

until $f(x+t\Delta x) < f(x) + \alpha t \nabla f(x)^T \Delta x$

reduce t. (backtrack) until $t < t_0$

i.e.



※ Generic descent method (key process)

进阶: Gradient Descent Method

convergence result: for strongly convex f

$$f(x^{(k)}) - p^* \le c^k (f(x^{(0)}) - p^*)$$

$(c \in (0,1)$ on m, $x^{(0)}$, line search type$)$

How to find c? $S = \{x \mid f(x) \le f(x^*)\}$

S is bounded

$\therefore$ This time $H(x) \triangleq \nabla^2 f(x)$

$\exists M > 0$, for any $x \in S^0$,

$|f(x)| \leq M$

$\nabla^2 f(x) \leq MI$

$\therefore \quad C = 1 - \frac{m}{M}$

(m: min eigenvalue)

(M: max eigenvalue)

if $C \to 1$: it is called "ill-condition"

~~th~~ that means $m \ll M$

, which causes the speed ~~to~~ becoming very slow.

Tutorial 8

Question 1

proof: $\because$ f is strongly convex

$\therefore \forall x, y \in S$

$f(y) \geq f(x) + \nabla f(x)^T (y-x) + \frac{m}{2} \|y-x\|_2^2$ (1)

let $g(y) =$ ~~RHS~~ RHS of (1)

$\bar{y} = x - \frac{1}{m} \nabla f(x)$ minimizes $g(y)$

$\therefore f(y) \geq f(x) \cancel{+} - \frac{1}{2m} \|\nabla f(x)\|^2$

$\therefore p^* \geq f(x) - \frac{1}{2m} \|\nabla f(x)\|^2$

Question 2

proof:

$\forall x \in S, \exists M > m > 0$

$mI \leq \nabla^2 f(x) \leq MI$

$x^+ = x - t\nabla f(x)$

$f(x^+) = f(x - t\nabla f(x))$

$= f(x) + \nabla f(x)^T [x^+ - x]$

$+ \frac{1}{2}[x^+ - x]^T H(\xi)[x^+ - x]$

Moreover $\because H(\xi) \leq MI$

$x^T - x = -t\nabla f(x)$

$\therefore f(x^+) \leq f(x) - t\|\nabla f(x)\|_2^2$

$+ \frac{Mt^2}{2}\|\nabla f(x)\|^2$

For any t,

$\cancel{\nabla f(t-t\Delta}$

$f(t - t\nabla f(x)) \leq$

$f(x) - t\|\nabla f(x)\|^2 +$

$\frac{Mt^2}{2}\|\nabla f(x)\|^2$

$\therefore \quad = f(x^*)$

When $t = \frac{1}{M}$:

$f(x^*) \leq$

$f(x) - \frac{1}{2M}\|\nabla f(x)\|^2$

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Question 2

proof: $\forall x \in S, \exists M > m > 0,$

$\therefore mI \le \nabla^2 f(x) = MI$

Moreover: $x^+ = \cancel{x} = x - t\nabla f(x)$

$\cancel{\therefore f(x^+) = f(x) - t\nabla f(x)}$ $f(x^+) = f(x - t\nabla f(x))$

$= f(x) + \nabla f(x)^T [x^+ - x]$

$+ \frac{1}{2}[x^+ - x]^T H(\xi)[x^+ - x]$

Moreover $\because H(\xi) \le mI$

$x^T - x = \cancel{-t\nabla f(x)}\ \cancel{-t\nabla f(x)}$

$-t\nabla f(x)$

$f(x^+ \le f(x) - t\|\nabla f(x)\|_2^2$

$+ \frac{Mt^2}{2}\|\nabla f(x)\|^2$

$\therefore$ for any t

$f(t - t\nabla f(x)$

$f(x) - t\|\nabla f(x)\|_2^2$

$+ \frac{Mt^2}{2}\|\nabla f(x)\|^2$

$= f(x^*)$

when $t = \frac{1}{m}$

$f(x^*) \le$

$f(x) - \frac{1}{2m}\|\nabla f(x)\|^2$

$$f(x^*) - p^* \le \left[f(x) - \cancel{p^*} - \frac{1}{2m}\|\nabla f(x)\|^2\right] - p^*$$

$$\therefore f(x^*) - p^* \le f(x) - p^* - \frac{1}{2m}\|\nabla f(x)\|^2$$

Moreover From Q1: $\|\nabla f(x)\|^2 \ge 2m[f(x) - p^*]$

$$\therefore f(x^*) - p^* \le \left[1 - \frac{m}{M}\right][f(x) - p^*]$$

Let $C = \left[1 - \frac{m}{M}\right]$

$$\therefore f(x^*) - p^* \le C^k [f(x^0) - p^*]$$

$$\{f(x^*) - p^* \le C[f(x) - p^*] \le C^k[f(x) - p^*]\}$$

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$$g'(y) = \nabla f(x)(-x) + m(y-x)(-x)$$

$\therefore$ 令 $g'(y) = 0$

$$\nabla f(x) + m(y-x) = 0$$

$$\therefore my = mx - \nabla f(x)$$

$$\therefore y = x - \frac{1}{m}\nabla f(x)$$

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Tutorial 9

Question 1

proof: $\because f$ is strongly convex.

$\therefore \forall x, y \in S$

$$f(y) \geq f(x) + \nabla f(x)^T (y-x) + \frac{m}{2} \|y-x\|^2 \quad (1)$$

let $g(y) =$ RHS of (1)

$\therefore \tilde{y} = x - \frac{1}{m} \nabla f(x)$ minimizes $g(y)$

$$\therefore f(y) \geq f(x) - \frac{1}{2m} \|\nabla f(x)\|^2$$

Moreover $\because f(y) = p^*$

$$\therefore p^* \geq f(x) - \frac{1}{2m} \|\nabla f(x)\|^2$$

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Question 2

proof: $\forall x \in S, \exists M > m > 0,$

$$mI \le \nabla^2 f(x) = mI$$

Moreover $\because x^+ = x - t\nabla f(x)$

$$\therefore f(x^+) = f(x - t\nabla f(x))$$

$$= f(x) + \nabla f(x)^T [x^+ - x]$$

$$+ \frac{1}{2}[x^+ - x]^T H(\xi)[x^+ - x]$$

Moreover $\because H(\xi) \le mI,$

$$x^T - x = -t\nabla f(x)$$

$$\therefore f(x^+) \le f(x) - t\|\nabla f(x)\|_2^2 + \frac{Mt^2}{2}\|\nabla f(x)\|^2$$

$\therefore$ For any t:

$$f(x^*) \le f(x^+) = f(x - t\nabla f(x))$$

$$\le f(x) - t\|\nabla f(x)\|_2^2 + \frac{Mt^2}{2}\|\nabla f(x)\|^2$$

~~$= f(x^*)$~~

when $t = \frac{1}{M}$

$f(x)$ reaches f_{max}

$$\therefore f(x^*) \le f(x) - \frac{1}{2M}\|\nabla f(x)\|^2$$

Now we have:

$$f(x^*) - p^* \le \left[f(x) - \frac{1}{2m}\|\nabla f(x)\|^2\right] - p^*$$

$$\therefore f(x^*) - p^* \le f(x) - p^* - \frac{1}{2m}\|\nabla f(x)\|^2$$

Moreover $\because$ From Q_1: $\|\nabla f(x)\|^2 \ge 2m[f(x) - p^*]$

$$\therefore f(x^*) - p^* \le \left[1 - \frac{m}{M}\right][f(x) - p^*]$$

Let $c = \left[1 - \frac{m}{M}\right]$

$$\therefore f(x^*) - p^* = c^k[f(x^0) - p^*]$$

$$\{f(x^*) - p^*) \le c[f(x) - p^*] \le c^k[f(x) - p^*]\}$$

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Question 3

For $k=0$, we can get the starting point $x^{(0)}=(\gamma,1)$

The gradient at $x^{(k)}$ is $(x_1^{(k)}, \gamma x_2^{(k)})$,

now we can get to verify $x^{(k-1)}$:

$$x^{(k+1)} = x^{(k)} - t\nabla f(x^k) = \begin{bmatrix} (1-t)x_1^{(k)} \\ (1-\gamma t)x_2^{(k)} \end{bmatrix}$$

$$= \left(\frac{\gamma-1}{\gamma+1}\right)^k \begin{bmatrix} (1-t)\gamma \\ (1-\gamma t)(-1)^k \end{bmatrix}$$

and

$$f(x^{(k)} - t\nabla f(x^{(k)})) = [\gamma^2(1-t)^2 + \gamma^2(1-\gamma^2 t)^2]\left(\frac{\gamma^2-1}{\gamma^2+1}\right)^{2k}$$

This is minimized by $t=\frac{2}{1+\gamma^2}$

$\therefore$ We have: $x^{(k+1)} = x^{(k)} - t\nabla f(x^{(k)})$

$$= \begin{bmatrix} (1-t)x_1^{(k)} \\ (1-\gamma^2 t)\gamma^2 x_2^{(k)} \end{bmatrix}$$

$$= \left(\frac{\gamma^2-1}{\gamma^2+1}\right)\begin{bmatrix} x_1^{(k)} \\ x_2^{(k)} \end{bmatrix}$$

$$= \left(\frac{\gamma-1}{\gamma+1}\right)^{k+1}\begin{bmatrix} \gamma^2 \\ (-1)^k \end{bmatrix}$$

From my own searches:

Tutorial 3:

For $k=0$, we get the starting point $x^{(0)} = (\gamma, 1)$

The gradient at $x^{(k)}$ is $(x_1^{(k)}, \gamma x_2^{(k)})$,

so we get

$$x^{(k)} - t\nabla f(x^{(k)}) \cancel{= (x_1^{(k)}, \gamma x_2^{(k)}) =}$$

$$= \begin{bmatrix} (1-t)x_1^{(k)} \\ (1-\gamma t)x_2^{(k)} \end{bmatrix} = \left(\frac{\gamma-1}{\gamma+1}\right)^k \begin{bmatrix} (1-t)\gamma \\ (1-\gamma t)(-1)^k \end{bmatrix}$$

and

$$f(x^{(k)} - t\nabla f(x^k)) = \left(\gamma^2(1-t)^2 + \gamma(1-\gamma t)^2\right)\left(\frac{\gamma-1}{\gamma+1}\right)^{2k}$$

This is minimized by $t = \frac{2}{1+\gamma}$, so we have:

$$x^{(k+1)} = x^{(k)} - t\nabla f(x^{(k)})$$

$$= \begin{bmatrix} (1-t)x_1^{(k)} \\ (1-\gamma t)x_2^{(k)} \end{bmatrix} \quad \therefore \text{Q.E.D.}$$

$$= \left(\frac{\gamma-1}{\gamma+1}\right)\begin{bmatrix} x_1^{(k)} \\ -x_2^{(k)} \end{bmatrix}$$

$$= \left(\frac{\gamma-1}{\gamma+1}\right)^{k+1}\begin{bmatrix} \gamma \\ (-1)^k \end{bmatrix}$$

Lecture 10 (Week 11)

Iterative Method

x : Now solution

x^+ : Next solution

$$x^+ = x + t\Delta x$$

t: step size (>0)

Δx: direction

Solution Step

(1) $\Delta x = ?$ direction $= ?$ → descent $f(x^+) \le f(x)$ $\Leftrightarrow \Delta x^T \nabla f(x) < 0$

(2) $t = ?$ stepsize $= ?$

exact linear search $t^* = \arg\min_{t>0} \{f(x + t\Delta x)\}$

backtracking linear search

$f(x + t\Delta x) < f(x) + \alpha t \nabla f(x)^T \Delta x$

$t = 1, \beta, \beta^2, \beta^3, \beta^4, \dots, \beta^n$

It obtained that

t satisfied $\beta t_0 \le t \le 1$

Revision of Lecture 9

⊗ x^* is optimal $\Leftrightarrow$ $\nabla f(x^*) = 0$

$\Updownarrow$

$f(x^*) = p^*$

$\nabla f(x) = Px + q$

OPT $\quad x = -\frac{q}{P}$

solution $\quad = -P^{-1}q$

$\therefore p^* = f(x^*)$

$= \frac{1}{2}[-P^{-1}q] \cancel{T} P [-P^{-1}q]$

$+ q[-P^{-1}q] + r$

Infeasible start Newton Method

primal - dual step:

$\nabla f(x+\Delta x_{nt}) \approx \nabla f(x) + \nabla^2 f(x)\Delta x_{nt}$

(given by $r(y)+Dr(y)\Delta y = 0$)

x is the current solution

$\Downarrow$

$y=(x,\mu)$ Primal-dual residual:

$\therefore r(y) = (\nabla f(x) + A^T\mu,\ Ax-b)$

Now $r(y) \neq 0$

Now we have $x^+ = \cancel{x+\Delta x_{nt}} = x+\Delta x_{nt}$

$\mu^+ = \mu + \Delta\mu_{nt}$

$y^+ = (x+\Delta x_{nt},\ \mu+\Delta\mu_{nt})$

$\therefore$ let $r(y^+)$ ~~=0~~ try to find $r(y^+) = 0$

$(\nabla f(x+\Delta x) + A^T(\mu+\Delta\mu),\ A(x+\Delta x)-b)$

$r(y) = 0$

if $\nabla f(x+\Delta x) + A^T\mu + \Delta\mu = 0$

and $A(x+\Delta x) - b = 0$

How to reduce or eliminate problem:

minimize problem: $\cancel{f(t+Z\hat{x})}$ $\cancel{f(t+z\hat{x})}$

$f(Fz+\hat{x})$

Newton Step:

$$x^+ = x + t\Delta x$$

current solution: x' is feasible

It means: $Ax' = b$

$$f(x) \approx f(x_0) + \nabla f(x_0)^T(x-x_0) + \frac{1}{2}(x-x_0)^T H(x_0)(x-x_0)$$

$Ax^+ = b$ $\quad$ $x^+ = x+v$

$\Updownarrow$

$Av = 0$ $\quad$ $A(x+v) = b$

$$\nabla f(x+v) + A^T w \approx \nabla f(x) + \nabla^2 f(x) v + A^T w = 0$$

$A(x+v) = b$

$\rightarrow Av = 0$

$\underbrace{\nabla f(x^*)}_{\text{non-linear}} + \underbrace{A^T \mu^*}_{\text{linear}} = 0$, $Ax^* = b$

$\min f(x)$
s.t. $Ax = b$

Optimality Solution
$Ax^* = b$
$\nabla f(x^*) + A^T \mu^* = 0$

$A : p \times n$
$x \in \mathbb{R}^n$
$b \in \mathbb{R}^p$
$\text{rank}(A) = p \leq n$
if $p = n$, only one possible/feasible solution, we don't need to make ~~optimaza~~ Optimaizition.
$\therefore$ usually $p < n$

If $f(x)$ is quadratic?
$\nabla f(x^*) = Px + q$

→ $\boxed{Ax^* = b, \quad Px^* + A^T \mu^* = -q}$ $\Leftrightarrow$ $\begin{bmatrix} \quad \end{bmatrix} \begin{bmatrix} x^* \\ \mu^* \end{bmatrix} = \begin{bmatrix} \quad \end{bmatrix}$

$\min f(x)$

s.t. $Ax = b$

$$A = \begin{pmatrix} a_1^T \\ a_2^T \\ a_3^T \\ a_4^T \\ \vdots \\ a_p^T \end{pmatrix} \qquad b = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ \vdots \\ b_p \end{pmatrix}$$

$$\because Ax = b \begin{cases} a_1^T x = b_1 \Leftrightarrow h_1(x) = 0, & h_1(x) = a_1^T x - b_1 \\ \vdots & \nabla h_1(x) = a_1 \\ a_p^T x = b_p \Leftrightarrow h_p(x) = 0, & h_p(x) = a_p^T x - b_p \\ & \nabla h_p(x) = a_p \end{cases}$$

$\min f(x)$

s.t. $h_i(x) = 0$

$i \in (1, p)$

KKT Condition: x^*, $\mu^* = (\mu_1^*, \mu_2^*, \mu_3^*, \dots, \mu_p^*)^T$

Condition (1) $Ax^* = b$

Condition 4 (2) $\nabla f(x^*) = -\sum_{i=1}^{p} \mu_i \nabla h_i(x^*)$

$$\Rightarrow \nabla f(x^*) + \sum_{i=1}^{p} \mu_i \nabla h_i(x^*) = 0$$

$$\Leftrightarrow \nabla f(x^*) + \sum_{i=1}^{p} \mu_i a_i^T = 0$$

$$\Leftrightarrow \nabla f(x^*) + \underbrace{\overbrace{\boxed{(a_1, \cdots, a_p)}}^{= A^T} \overbrace{\boxed{\begin{pmatrix} \mu_1 \\ \vdots \\ \mu_p \end{pmatrix}}}^{= \mu^*}}_{\text{(Matrix Form)}} = 0 \Leftrightarrow \nabla f(x^*) + A^T \mu^* = 0$$

Tutorial 10

Question 1

Suppose x is the current solution. ~~Δx~~

$\therefore$ Now Δx is = descent direction

~~$f(x = t + \Delta x$~~

$$f(x+t\Delta x) = f(x) + \nabla f(x)^T (t\Delta x)$$

(Moreover $\because$)

$$+ \frac{1}{2} [\cancel{t\Delta}\, t\Delta x]^T H(\xi) [t\Delta x] \quad ①$$

& $\because$ $H(\xi) \leq \cancel{mI}\ MI$

$$f(x+t\Delta x) \leq f(x) + [\nabla f(x)^T \Delta x] t + \frac{M}{2} \|\Delta x\|^2 t^2 \quad ②$$

Stopping Condition in Backtracking Line Search:

$$f(x+t\Delta x) \leq f(x) + \alpha [f(x)^T \Delta x] t \quad ③$$

When RHS of ② $\leq$ RHS of ③

the stopping condition ~~get~~ met. ④

⑤

④ $\Leftrightarrow$ $t(1-\alpha) \nabla f(x)^T \Delta x + \frac{M}{2} t^2 \|\Delta x\|^2 \leq 0$

when $0 < \alpha < 1$, fixed.

⑤ $\Rightarrow$ $t \leq 2(1-\alpha) \cancel{\frac{\nabla f(x) \Delta x}{\|m\|}} \dfrac{\nabla f(x) \Delta x}{M \|\Delta x\|^2}$

In backtracking L.S.,

$$\beta^{k} \le -2(1-\alpha) \frac{\nabla f(x)^{T} \Delta x}{M \|\Delta x\|^{2}}$$

$$\text{to}^{a}\ k \le \log_{\beta}\left[-2(1-\alpha) \cdot \frac{\nabla f(x)^{T} \Delta x}{M \|\Delta x\|^{2}}\right]$$

Newton's Method

③ $\Delta x_{nd} = -\nabla^2 f(x)\nabla f(x)$

$= -H(x)^{-1}\nabla f(x)$

$\hat{f}\ \cancel{(x+v)}$

$\min f(x) \Longleftrightarrow \nabla f(x) = 0$

$f(x) - \inf_y \hat{f}(y) = \frac{1}{2}\lambda(x)^2$

实际问题

$f(x) \rightarrow y = Ax$

$f(A^{-1}y)$

Classical convergence analysis

步长小

误差率快速减小

$\exists\, \eta > 0,\ \cancel{P \geq 0}\ \gamma > 0$

Phase 1:

$\|\nabla f(x)\| \geq \eta$

$x_0, x_1, \ldots\ldots, x^k$

$f(x^i) - f(x^{i+1}) < \gamma$

步长大

误差率减少的速度慢

Phase 2:

$\|\nabla f(x)\| < \eta$

$f(x^{k+1}) - p$

$\cancel{\leq (\ \ p)^2}$

$\leq \cancel{(\eta)^2}\ (\cancel{\ \ }\ \eta)^t$

(~~累积~~ 累积最佳)

η：误差率

t：迭代次数 ~~(偏差)~~

$$\therefore \|x\|_P = \sqrt{x^T P x}$$
$$= \sqrt{x^T P^{\frac{1}{2}} P^{\frac{1}{2}} x} \quad [(P^{\frac{1}{2}})^T = P^{\frac{1}{2}}]$$
$$= \sqrt{(P^{\frac{1}{2}}x)^T \cdot (P^{\frac{1}{2}}x)}$$
$$= \| P^{\frac{1}{2}} x \|_2$$

Now we proved $\|x\|_P = \|P^{\frac{1}{2}}x\|_2$

$$\cancel{\|x\|_* = \max\{x^T x\}}$$

$$\cancel{\|y\|_* = \max\{x^T y \mid \|x\|_P = 1\}}$$
$$= \cancel{\sqrt{x^T P^T x}}$$
$$= \cancel{\sqrt{y^T P^{-1} y}}$$

$$\cancel{\|y\|_* = \sqrt{}}$$

$$\|y\|_* = \max\{x^T y \mid \|x\|_P = 1\}$$
$$= \sqrt{y^T P^{-1} y}$$
$$= \|P^{-\frac{1}{2}} y\|_2$$

Now if we use $\|\cdot\| = \|\cdot\|_P$

$$\Delta x_{sd} = -P^{-1} \nabla f(x)$$

lecture Week 10

Review

$p^* = f(x^*) \Leftrightarrow x^*$ is optimal

$\Leftrightarrow \nabla f(x^*) = 0$

$x^0 \to x^1 \to \cdots \to x^n$

Initial Point and sublevel set

S is the set.

We can assume that

$g(x) = f(x) - \frac{m x^T x}{2}$

$\nabla^2 g(x) = \nabla^2 f(x) - mI$

Error : $f(x) - p^* \leq \frac{1}{2m} \|\nabla f(x)\|_2^2$

(≥ 0)

Generic Descent Methods

$f(x^{k+1}) < f(x^k)$

$x^{(k+1)} = x^{(k)} + t^{(k)} \Delta x^{(k)}$

$x^+ = x + \underbrace{t}_{\text{Step size } (t>0)} \underbrace{\Delta x}_{\text{search direction}}$

steepest

Quadratic function on R^2

if $\frac{m}{M} \ll 1$, the Gradient Descent Function will not work well. in direct use.

x: current solution

$x^+ = x + V$

But how we can find v?

$f(x+V) \doteq f(x) + \nabla f(x)^T V$

We suppose $\|V\| \leq 1$

we hope $f(x+V)$ as small as possible.

$\Delta X_{nsd} = \arg\min \nabla f(x)^T V \quad (\|v\| < 1)$

(normal search direction)

$\Leftrightarrow$ ΔX_{nsd} is the ~~opti~~ optimal solution

to $\min \nabla f(x)^T V \quad (<0)$

s.t. $\|V\| \leq 1 \Rightarrow (\|V\| = 1)$

$\therefore \|\Delta_{snd}\| = 1$

① $\therefore \Delta x^{\dagger}$ ~~且~~ : $\Delta x^{T} \nabla f(x) < 0$

↓ directional derivates along Δx

~~沿~~ Δx

② t { excat LS ~~(Newton's Method)~~ (Costy)
Backtrack

Try 1 : $t = 1$

check $f(x + t\Delta x) < f(x) + \alpha t \nabla f(x)^{T} \Delta x$

Try 2 : $t = \beta$

Try 3 : $t = \beta^2$

......

Gradient Descent

① $\Delta x = ~~\text{方}~~ -\nabla f(x)$

$c = 1 - \frac{m}{M}$

For all $x \in S$,

$H(x) \leq ~~mI~~ MI$

For $c = 1 - \frac{m}{M}$:

For all $x \in S$

$mI \leq H(x) \leq MI$

$1 - \frac{m}{M} \leq \varepsilon$

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$\|\Delta_{sgd}\| = 1$

Legth of $\nabla f(x)$

~~Length of $\nabla f(x)$~~

$\|\cdot\|$ is a norm

The dual norm of its is :

$$\|a\|_* = \max_{\|x\|=1} \{x^T a\}$$

Steepest descent Method

① $\Delta x = -\nabla f(x)$

② $\Delta x = \Delta x_{sd} = \|\nabla f(x)\|_* \cdot \Delta x_{nsd}$

$$\Delta x_{nsd} = \underset{\|v\|=1}{\operatorname{argmin}} \{\nabla f(x)^T v\}$$

$$\begin{cases} \|\cdot\| = \|\cdot\|_2 \\ \Delta x_{sd} = -\nabla f(x) \end{cases}$$

$P > 0$, $x \in R^n$

$\|x\|_P = \sqrt{x^T P x} > 0$ if $x \neq 0$

$(\sqrt{x^T P x} = 0$ if $x = 0)$

$P > 0$

$P = U D U^T$

$P^{\frac{1}{2}} = U \sqrt{D} U^T$

$P = P^{\frac{1}{2}} \cdot P^{\frac{1}{2}}$

$= \sqrt{x^T P^{\frac{1}{2}} P^{\frac{1}{2}} x}$

$\cancel{(P^{\frac{1}{2}})^T = P^{\frac{1}{2}}}$

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③ $\Delta x = -\nabla f(x)$ Steepest Descent Method

$1 - \frac{m}{M}$ ~~≈ 1~~ $\ll 1$ and very slow.

④ $\Delta x_{nsd} = \cancel{\arg\min \{\nabla f(x)^T v\}}$
$= \arg\min \{ \nabla f(x)^T v \mid u + v = 1 \}$

$\Delta x = \Delta_{sd} = \|\nabla f(x)\|_* \cdot \Delta_{nsd}$

$\cancel{\Delta x_{sd} = \nabla^2 f(x)}$
$\cancel{= [\nabla^2 f(x)]^{-1} \nabla f(x)}$

$\cancel{\nabla^2 f(x)^{-1}}$

⑤ $\Delta x_{nt} = -[\nabla^2 f(x)]^{-1} \cdot \nabla f(x)$
Newton's Method

Lecture Week 12

Revision

@ KKT condition

$\min f(x)$
s.t. $Ax=b$

f is convex
$A \in R^{p\times n}$ with rank $A = p$
p^* is finite and attained

optimality problems

$$\nabla f(x^*) + A^T\mu^* = 0 \quad , \quad Ax^* = b$$

$\therefore$

To find
$\min f(x)$ $\Leftrightarrow$
$Ax=b$

To find x^*, μ^*
such that
$\nabla f(x^*) + A^T\mu^* = 0$
$Ax^* = b$

New to learn: Inequality constrained minimization

minimize $f_0(x)$
subject to $f_i(x) \le 0 \ (i \in (1,m))$
$Ax=b$

$f_0(x^*) = p^*$ strong: $p^* = d^*$ weak: $p^* \geq d^*$

Example: LP. QP. QCQP. GP

How to solve those problems?

Problem A

Logarithmic Barrier

$\min f_0(x)$

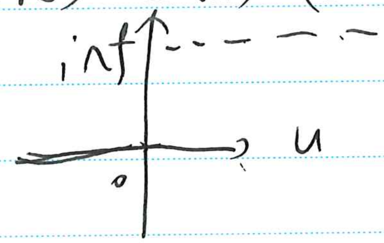
s.t. $f_1(x) \leq 0$

$f_2(x) \leq 0$

$Ax = b$

$$\cancel{I_-(x)} \quad I_-(u) = \begin{cases} 0 & u \leq 0 \\ +\infty & u > 0 \end{cases}$$

$(u \in R)$ $I_-(u)$ (Otherwise)



Actually, A $\Leftrightarrow$ $\min\limits_{x} f_0(x) + I^-(f_1(x)) + I^-(f_2(x))$

s.t. $Ax = b$

x^* is optimal solution to A^*

$\Leftrightarrow$ x^* is optimal solution to B^*

$$I^-(u) \triangleq -\frac{1}{t} \ln(-u)$$

$-\frac{1}{t} \ln(-u)$

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Central Path is $\{x^*(t) \mid t>0\}$

$$\min \ t \cdot f_0(x) + \Phi(x)$$
$$\text{s.t. } Ax = b$$

$x^*(t)$'s KKT condition:

1. $Ax^*(t) = b$
2. $t\nabla f_0(x^*(t)) + \nabla\Phi(x^*(t)) + A^T w = 0,\ Ax = b$

$\cancel{=f}$

Lagrangian Function Revision

$\min f_0(x)$ $\quad \therefore L_1(x,\lambda,\mu) = f_0(x) + \sum \lambda_i(x) f_i(x)$

s.t. $f_i(x) \le 0$

$h_i(x) = 0$ $\quad + \sum \mu_i(x) h_i(x)$

$$\therefore L(x, \lambda^*(t), v^*(t)) = f_0(x) + \sum_{i=1}^{m} \lambda_i^*(t) f_i(x) + v^*(t)^T (Ax-b)$$

$\cancel{+\sum}$

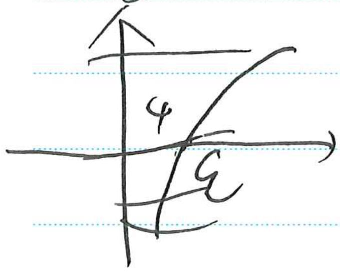
where we define $\lambda_i^*(t) = -\frac{1}{t f_i(x^*(t))}$

$v^*(t) = \frac{w}{t}$

$$p^* \ge g(\lambda^*(t), v^*(t)) = L(x^*(t), \lambda^*(t), v^*(t))$$
$$= f_0(x^*(t)) - \frac{m}{t}$$

图:

~~fig. y=loll~~



~~$L_0 = 0 \quad L_0 \quad Y_u$~~

$$\Rightarrow \min f_0(x) - \frac{1}{t}\ln[-f_1(x)] - \frac{1}{t}\ln[-f_2(x)]$$

$$\text{s.t. } Ax = b$$

$$\Downarrow$$

$$\min f_0(x) - \frac{1}{t}\sum_{i=1}^{m} \ln[f_i(x)]$$

$$\text{s.t. } Ax = b$$

$$\Downarrow$$

$$\min f_0(x) + \frac{1}{t}\sum_{i=1}^{m} -\ln[f_i(x)]$$

$$\text{s.t. } Ax = b$$

We define $\sum_{i=1}^{m} -\ln[-f_i(x)] = \Phi(x)$

$$\operatorname{dom}\phi = \{x \mid f_1(x) < 0, \ldots, f_m(x) < 0\}$$

$$\therefore \nabla\phi(x) = \sum_{i=1}^{m} \frac{1}{-f_i(x)} \nabla f_i(x)$$

$$\therefore \nabla^2\Phi(x) = \sum_{i=1}^{m} \frac{1}{f_i(x)^2} \nabla f_i(x) \nabla f_i(x)^T + \sum_{i=1}^{m} \frac{1}{-f_i(x)} \nabla^2 f_i(x)$$

$\nabla^2\Phi(x)$ is semi-positive $\therefore \Phi(x)$ convex ※

(半正定知识相关)

$A \in S^n$ (对称矩阵)

$A \in S_+^n \Leftrightarrow \forall x \in R^n$, $x^T A x \geq 0$

$Q = \alpha\alpha^T$

Proof: $\forall x \in R^n$

$\cancel{x^T A}\, x^T Q x =$

$x^T[\alpha\alpha^T]x$

$= x^T\alpha\alpha^T x$

$= (\alpha^T x)^2 \geq 0$

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$\delta x_{nsd} = \Delta x_{nd}$

$\Delta x_{nsd} = \arg\min\{v^T \nabla f(x) \mid \|v\| = 1\}$

$\Updownarrow$

Δx_{nsd} is the optimal solution to

$$\begin{aligned} \min \quad & v^T \nabla f(x) \\ \text{s.t.} \quad & \|v\| = 1 \end{aligned}$$

$\Delta x_{sd} \triangleq \|\nabla f(x)\|_{*} \cdot \Delta x_{nsd}$

$\|\cdot\|$ is given

$v \in \mathbb{R}^n \quad \|v\|_* \triangleq \underset{\max}{\sup}\{v^T x \mid \|x\| = 1\}$

$\max\{v^T x \mid \|x\| = 1\} = a$

$\min\{v^T x \mid \|x\| = 1\} = -a$

互为 ~~相关数~~

互为相反数

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Tutorial 11

Question 1

Solve: We can introduce slack variables $s \in R^n$ such that $Ax+s=b$ and $s>0$.

$\therefore$ We can get logarithmic barrier function:

$$\phi(x) = -\mu \sum_{i=1}^{n} \log_e(s_i) = -\mu \sum_{i=1}^{n} \log_e(b_i - A_i x)$$

$\therefore$ ~~Now we can ge~~

$\therefore$ Now we can get unconstrained problem:

$$f(x) = \min_x \left(c^T x - \mu \sum_{i=1}^{n} \cancel{\log_e(b_i - A_i x)} \ln(b_i - A_i x) \right)$$

Moreover $\because$ $\nabla f(x) = c - \mu A^T \left[\frac{1}{b_1 - A_1 x}, \frac{1}{b_2 - A_2 x}, \cdots\cdots, \frac{1}{b_n - A_n x} \right]^T$

$$\therefore H(x) = \nabla^2 f(x) = \mu A^T \begin{bmatrix} \frac{1}{(b_1 - A_1 x)^2} & 0 & \cdots\cdots & 0 \\ 0 & \frac{1}{(b_2 - A_2 x)^2} & \cdots & 0 \\ \vdots & & \ddots & \\ 0 & \cdots\cdots & \cdots & \frac{1}{(b_n - A_n x)^2} \end{bmatrix} A$$

$\therefore$ $\nabla^2 f(x) \Delta x = -\nabla f(x)$

$\therefore$ Now we can do line search, we choose x_0 strictly feasible (i.e., $Ax_0 < b$), $\mu_0 > 0$, and tolerance ε.

Then we ~~inte~~ iterate by using Newton's Method:

Update $\mu_{k+1} = \sigma \mu_k$

Terminate when $\mu_k < \varepsilon$

Problem 2

solve: Considering this, we can introduce

$$\phi(x) = -\mu \sum_{i=1}^{n} [\log_e(x_i) + \log_e(1-x_i)]$$

$\therefore$ We have an unconstrained problem now:

$$\min_{x} \left[\tfrac{1}{2}(x-a)^T P^{-1}(x-a) \right)$$

Moreover $\because \nabla f(x) = P^{-1}(x-a) - \mu\left[\frac{1}{x_1} - \frac{1}{1-x_1}, \cdots\cdots, \frac{1}{x_n} - \frac{1}{1-x_n}\right]^T$

$$\therefore \nabla^2 f(x) = H(\xi) = P^{-1} + \mu \begin{bmatrix} \frac{1}{x_1^2} + \frac{1}{(1-x_1)^2} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \frac{1}{x_n^2} + \frac{1}{(1-x_n)^2} \end{bmatrix}$$

$\therefore$ When we try to get the solution of Δx,

we find: $(P^{-1} + \mu D)\Delta x = -\nabla f(x)$

where $D = \mathrm{diag}\left(\frac{1}{x_i^2} + \frac{1}{(1-x_i)^2}\right) \quad (i \in [\cancel{0}1, n])$

$\therefore$ As the same method, we use Newton's Method:

We choose x_0 strictly feasible (i.e. $0 < x_i < 1$),

$\mu_0 > 0$, tolerance ε

$\therefore$ We now iterate, make $\mu_{k+1} = \sigma\mu_k$,

terminate when $\mu_k < \varepsilon$

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~~Problem 2~~ ~~题~~ solve. According to the previous proof.

~~solve:~~ $\nabla f(x) = (x_1, \gamma x_2)$ we can know:

$$\alpha = \frac{2}{\gamma+1}$$

$$x_1^{(k+1)} = x_1^{(k)} \cdot \frac{\gamma-1}{\gamma+1}, \quad x_2^{(k+1)} = x_2^{(k)} \left(-\frac{\gamma-1}{\gamma+1}\right)$$

∴ Inductively, the close form expressions hold.

∴ when we apply Netwon Method

$$H = H(g) = \begin{bmatrix} 1 & 0 \\ 0 & \gamma \end{bmatrix} \quad \therefore H^{-1}(g) = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{\gamma} \end{bmatrix}$$

Now we let

$$\Delta x = -H^{-1} \nabla f(x) = \cancel{(\gamma,}$$

$$dx^{(k)} = x^{k} + t\Delta x^{k+1} = x$$

$$\therefore x^{(1)} = x^{(0)} + \alpha d$$

$$= (\gamma, 1) - (\gamma, \frac{\gamma}{\gamma})$$

$$= (\gamma, 1) - (\gamma, -1) = (0, 0)$$

$$\therefore x^{(k)} = (0, 0)$$

∴ ~~sloved~~ solved.